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Substrate processing apparatus, furnace opening assembly, substrate processing method, method of manufacturing semiconductor device and non-transitory tangible medium

Abstract

There is provided a technique capable of reducing a corrosion of a furnace opening shutter. According to one aspect thereof, a substrate processing apparatus includes: a process vessel provided with an opening and a first sealing surface around the opening; and a furnace opening shutter provided with a second sealing surface facing the first sealing surface, and capable of closing the opening. The furnace opening shutter includes: a protective cover provided corresponding to an inner surface of the furnace opening shutter, facing the process vessel and greater than the opening; and a lid provided corresponding to an outer surface of the furnace opening shutter and supporting the protective cover. The second sealing surface is configured by an outer peripheral portion of the protective cover and the lid, and a purge gas is supplied to a gap between the first sealing surface and the second sealing surface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0252434	12/2012	Yuasa et al.	N/A	N/A
2015/0267296	12/2014	Nishida et al.	N/A	N/A
2015/0279712	12/2014	Yachi	438/795	H01L 21/67196
2016/0362784	12/2015	Isobe et al.	N/A	N/A
2017/0335452	12/2016	Yamazaki et al.	N/A	N/A
2020/0066551	12/2019	Okajima et al.	N/A	N/A
2020/0149159	12/2019	Okajima et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103985632	12/2013	CN	H01L 21/6719
2002-009010	12/2001	JP	N/A
2002025911	12/2001	JP	N/A
2007-073746	12/2006	JP	N/A
2008-078285	12/2007	JP	N/A
2010-021385	12/2009	JP	N/A
2011-171657	12/2010	JP	N/A
2015-185662	12/2014	JP	N/A
2017-005090	12/2016	JP	N/A
2018-022773	12/2017	JP	N/A
2020-035779	12/2019	JP	N/A
10-2020-0094893	12/2019	KR	N/A
102447175	12/2019	KR	N/A

OTHER PUBLICATIONS

Singapore Written Opinion issued on Jul. 11, 2023 for Singapore Patent Application No.

10202201914S. cited by applicant

Taiwan Office Action issued on May 2, 2023 for Taiwan Patent Application No. 111105421. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This non-provisional U.S. patent application claims priority under 35 U.S.C. § 119 of Japanese Patent Application No. 2021-046756, filed on Mar. 22, 2021, in the Japanese Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Field

(2) The present disclosure relates to a substrate processing apparatus, a furnace opening assembly, a substrate processing method, a method of manufacturing a semiconductor device and a non-transitory tangible medium.

2. Related Art

(3) As a part of a manufacturing process of a semiconductor device, for example, a film-forming step of forming a film on a substrate such as a semiconductor wafer may be performed. The film-forming step is performed by supplying a process gas to a process chamber in which the substrate is accommodated. The film-forming step is performed to form the film on a surface of the substrate. However, in reality, deposits containing a substance of the film may adhere to a location such as an inner wall of the process chamber as well as the surface of the substrate. The deposits may accumulatively adhere to the location whenever the film-forming step is performed. When a thickness of the deposits reaches a certain thickness or more, the deposits may be peeled off from the location such as the inner wall of the process chamber. As a result, foreign substances (particles) may be generated in the process chamber. Therefore, whenever the thickness of the deposits reaches the certain thickness, it is preferable to clean an inside of the process chamber or components in the process chamber by removing the deposits. As a method of removing the deposits, a dry cleaning process is disclosed in which an empty boat is transferred (loaded) into the process chamber and an etching gas (which is a cleaning gas) is supplied to remove the deposits by a dry etching process.

(4) However, the boat may not be cleaned. Therefore, a cleaning process may be performed by using a furnace opening shutter which is normally closed to maintain an inside of a furnace warm and to shield a transfer chamber from a heat when the boat is lowered. However, since the cleaning process uses a gas highly corrosive to metal, the furnace opening shutter may be corroded.

SUMMARY

(5) According to the present disclosure, there is provided a technique capable of reducing a corrosion of a furnace opening shutter.

(6) According to one or more embodiments of the present disclosure, there is provided a technique related to a substrate processing apparatus including: a process vessel provided with an opening through which a substrate retainer accommodating a substrate is transferred and a first sealing surface around the opening; and a furnace opening shutter provided with a second sealing surface

facing the first sealing surface, and capable of closing the opening in a state where the substrate retainer is not accommodated in the process vessel, wherein the furnace opening shutter includes: a protective cover provided corresponding to an inner surface of the furnace opening shutter, facing the process vessel and greater than the opening; and a lid provided corresponding to an outer surface of the furnace opening shutter and supporting the protective cover, wherein the second sealing surface is configured by an outer peripheral portion of the protective cover and the lid, and a purge gas is supplied to a gap between the first sealing surface and the second sealing surface from a location radially more outward than the outer peripheral portion of the protective cover.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram schematically illustrating a vertical cross-section of a vertical type process furnace of a substrate processing apparatus preferably used in one or more embodiment of the present disclosure, wherein a process furnace **202** is schematically illustrated.
- (2) FIG. 2 is a diagram schematically illustrating a horizontal cross-section taken along the line A-A of the process furnace **202** of the substrate processing apparatus shown in FIG. 1 preferably used in the embodiments of the present disclosure
- (3) FIG. 3 is a block diagram schematically illustrating a configuration of a controller **121** and related components of the substrate processing apparatus preferably used in the embodiments of the present disclosure.
- (4) FIG. 4 is a diagram schematically illustrating a configuration of a process vessel of the substrate processing apparatus preferably used in the embodiments of the present disclosure, wherein a cross-section of a furnace opening shutter **219s** is schematically illustrated.
- (5) FIG. 5 is a diagram schematically illustrating a configuration of the furnace opening shutter **219s** of the substrate processing apparatus preferably used in the embodiments of the present disclosure, wherein a top perspective view of the furnace opening shutter **219s** is schematically illustrated.
- (6) FIG. 6 is a diagram schematically illustrating a configuration of a cooling water system of the substrate processing apparatus preferably used in the embodiments of the present disclosure, wherein a portion of the process furnace **202** is schematically illustrated.
- (7) FIG. 7 is a flowchart schematically illustrating a method of manufacturing a semiconductor device according to the embodiments of the present disclosure.

DETAILED DESCRIPTION

Embodiments

(8) Hereinafter, one or more embodiments (also simply referred to as “embodiments”) according to the technique of the present disclosure will be mainly described with reference to the drawings. The drawings used in the following descriptions are all schematic. For example, a relationship between dimensions of each component and a ratio of each component shown in the drawing may not always match the actual ones. Further, even between the drawings, the relationship between the dimensions of each component and the ratio of each component may not always match.

(9) (1) Configuration of Substrate Processing Apparatus

(10) As shown in FIG. 1, a substrate processing apparatus according to the present embodiments includes a process furnace **202**. The process furnace **202** is provided with a heater **207** serving as a temperature regulator (which is a heating structure or a heating system). The heater **207** is of a cylindrical shape, and is vertically installed while being supported by a support plate (not shown). The heater **207** also functions as an activator (also referred to as an “exciter”) capable of activating (exciting) a gas such as a process gas by a heat.

(11) A reaction tube **203** is provided in an inner side of the heater **207** to be aligned in a manner

concentric with the heater **207**. For example, the reaction tube **203** is made of a heat resistant material such as quartz (SiO₂) and silicon carbide (SiC). The reaction tube **203** is of a cylindrical shape with a closed upper end and an open lower end. A manifold **209** is provided under the reaction tube **203** to be aligned in a manner concentric with the reaction tube **203**. For example, the manifold **209** is made of a metal such as stainless steel (SUS). The manifold **209** is of a cylindrical shape with open upper and lower ends. The upper end of the manifold **209** is engaged with the lower end of the reaction tube **203** so as to support the reaction tube **203**. An O-ring **220a** serving as a seal is provided between the manifold **209** and the reaction tube **203**. Similar to the heater **207**, the reaction tube **203** is installed vertically. A process vessel (also referred to as a “reaction vessel”) is constituted mainly by the reaction tube **203** and the manifold **209**. A process chamber **201** is provided in a hollow cylindrical portion of the process vessel. The process chamber **201** is configured to accommodate a plurality of wafers including a wafer **200** serving as a substrate. Hereinafter, the plurality of wafers including the wafer **200** may also be simply referred to as “wafers **200**”. The wafers **200** are processed in the process chamber **201**.

(12) Nozzles **249a**, **249b** and **249c** are provided in the process chamber **201** so as to penetrate a side wall of the manifold **209**. The nozzles **249a**, **249b** and **249c** serve as a first supplier (which is a first supply structure), a second supplier (which is a second supply structure) and a third supplier (which is a third supply structure), respectively. The nozzles **249a**, **249b** and **249c** may also be referred to as a first nozzle, a second nozzle and a third nozzle, respectively. For example, each of the nozzles **249a**, **249b** and **249c** is made of a heat resistant material such as quartz and silicon carbide. Gas supply pipes **232a**, **232b** and **232c** are connected to the nozzles **249a**, **249b** and **249c**, respectively. The nozzles **249a**, **249b** and **249c** are different nozzles, and the nozzles **249a** and **249b** are provided adjacent to each other.

(13) Mass flow controllers (also simply referred to as “MFCs”) **241a**, **241b** and **241c** serving as flow rate controllers (flow rate control structures) and valves **243a**, **243b** and **243c** serving as opening/closing valves are sequentially installed at the gas supply pipes **232a**, **232b** and **232c**, respectively, in this order from upstream sides to downstream sides of the gas supply pipes **232a**, **232b** and **232c** in a gas flow direction. Gas supply pipes **232d** and **232e** are connected to the gas supply pipe **232a** at a downstream side of the valve **243a** of the gas supply pipe **232a**. Gas supply pipes **232f** and **232h** are connected to the gas supply pipe **232b** at a downstream side of the valve **243b** of the gas supply pipe **232b**. A gas supply pipe **232g** is connected to the gas supply pipe **232c** at a downstream side of the valve **243c** of the gas supply pipe **232c**. MFCs **241d**, **241e**, **241f**, **241g** and **241h** and valves **243d**, **243e**, **243f**, **243g** and **243h** are sequentially installed at the gas supply pipes **232d**, **232e**, **232f**, **232g** and **232h**, respectively, in this order from upstream sides to downstream sides of the gas supply pipes **232d**, **232e**, **232f**, **232g** and **232h** in the gas flow direction. Each of the gas supply pipes **232a** through **232h** is made of a metal material such as stainless steel (SUS).

(14) As shown in FIG. 2, each of the nozzles **249a** and **249b** is installed in an annular space provided between an inner wall of the reaction tube **203** and the wafers **200** when viewed from above, and extends upward from a lower portion toward an upper portion of the reaction tube **203** along the inner wall of the reaction tube **203** (that is, extends upward along a wafer arrangement direction). That is, each of the nozzles **249a** and **249b** is installed in a region that is located beside and horizontally surrounds a wafer arrangement region in which the wafers **200** are arranged (stacked) so as to extend along the wafer arrangement region. Unlike the nozzles **249a** and **249b**, the nozzle **249c** extends in a horizontal direction without extending upward. When viewed from above, the nozzle **249b** is arranged so as to face an exhaust port **231a** described later along a straight line (denoted by “L” shown in FIG. 2) with a center of the wafer **200** transferred (loaded) into the process chamber **201** interposed therebetween. The straight line L is a straight line passing through the nozzle **249b** and a center of the exhaust port **231a**. When viewed from above, the nozzles **249a** and **249c** are arranged along the inner wall of the reaction tube **203** (an outer

peripheral portion of the wafer **200**) such that the straight line L is interposed therebetween. The straight line L is also a straight line passing through the nozzle **249b** and the center of the wafer **200**. A plurality of gas supply holes **250a**, a plurality of gas supply holes **250b** and a gas supply hole **250c** are provided at side surfaces of the nozzles **249a**, **249b** and **249c**, respectively. Gases are supplied through the gas supply holes **250a**, the gas supply holes **250b** and the gas supply hole **250c**. The gas supply holes **250a** and the gas supply holes **250b** are open toward the exhaust port **231a** when viewed from above, and are configured such that the gases are supplied toward the wafer **200** through the gas supply holes **250a** and the gas supply holes **250b**. The gas supply holes **250a** and the gas supply holes **250b** are provided from the lower portion toward the upper portion of the reaction tube **203**. The gas supply hole **250c** is provided toward a circumferential direction.

(15) A reactive gas is supplied into the process chamber **201** through the gas supply pipe **232a** provided with the MFC **241a** and the valve **243a** and the nozzle **249a**.

(16) A source gas is supplied into the process chamber **201** through the gas supply pipe **232b** provided with the MFC **241b** and the valve **243b** and the nozzle **249b**.

(17) A cleaning gas is supplied into the process chamber **201** through the gas supply pipe **232c** provided with the MFC **241c** and the valve **243c** and the nozzle **249c**.

(18) The cleaning gas is supplied into the process chamber **201** through the gas supply pipe **232d** provided with the MFC **241d** and the valve **243d**, the gas supply pipe **232a** and the nozzle **249a**.

(19) An inert gas is supplied into the process chamber **201** through the gas supply pipes **232e**, **232f** and **232g** provided with the MFCs **241e**, **241f** and **241g** and the valves **243e**, **243f** and **243g**, respectively, the gas supply pipes **232a**, **232b** and **232c** and the nozzles **249a**, **249b** and **249c**.

(20) A silicon-containing substance or a nitrogen-containing substance is supplied into the process chamber **201** through the gas supply pipe **232h** provided with the MFC **241h** and the valve **243h**, the gas supply pipe **232b** and the nozzle **249b**.

(21) A reactive gas supplier (which is a reactive gas supply structure or a reactive gas supply system) is constituted mainly by the gas supply pipe **232a**, the MFC **241a** and the valve **243a**. A source gas supplier (which is a source gas supply structure or a source gas supply system) is constituted mainly by the gas supply pipe **232b**, the MFC **241b** and the valve **243b**. A first cleaning gas supplier (which is a first cleaning gas supply structure or a first cleaning gas supply system) is constituted mainly by the gas supply pipe **232c**, the MFC **241c** and the valve **243c**. A second cleaning gas supplier (which is a second cleaning gas supply structure or a second cleaning gas supply system) is constituted mainly by the gas supply pipe **232d**, the MFC **241d** and the valve **243d**. An inert gas supplier (which is an inert gas supply structure or an inert gas supply system) is constituted mainly by the gas supply pipes **232e**, **232f** and **232g**, the MFCs **241e**, **241f** and **241g** and the valves **243e**, **243f** and **243g**. A silicon-containing substance supplier (which is a silicon-containing substance supply structure or a silicon-containing substance supply system) or a nitrogen-containing substance supplier (which is a nitrogen-containing substance supply structure or a nitrogen-containing substance supply system) is constituted mainly by the gas supply pipe **232h**, the MFC **241h** and the valve **243h**.

(22) In the present embodiments, the source gas and the reactive gas serve as a film-forming gas (that is, the process gas). Therefore, the source gas supplier and the reactive gas supplier may also be collectively or individually referred to as a “film-forming gas supplier” which is a film-forming gas supply structure or a film-forming gas supply system.

(23) Any one or an entirety of the gas suppliers described above may be embodied as an integrated gas supply system **248** in which the components such as the valves **243a** through **243h** and the MFCs **241a** through **241h** are integrated. The integrated gas supply system **248** is connected to the respective gas supply pipes **232a** through **232h**. An operation of the integrated gas supply system **248** to supply various gases to the gas supply pipes **232a** through **232h**, for example, operations such as an operation of opening and closing the valves **243a** through **243h** and an operation of adjusting flow rates of the gases through the MFCs **241a** through **241h** may be controlled by a

controller **121** which will be described later. The integrated gas supply system **248** may be embodied as an integrated structure (integrated unit) of an all-in-one type or a divided type. The integrated gas supply system **248** may be attached to or detached from the components such as the gas supply pipes **232a** through **232h** on a basis of the integrated structure. Operations such as maintenance, replacement and addition of the integrated gas supply system **248** may be performed on a basis of the integrated structure.

(24) The exhaust port **231a** through which an inner atmosphere of the process chamber **201** is exhausted is provided at a lower side wall of the reaction tube **203**. As shown in FIG. 2, the exhaust port **231a** is arranged at a location so as to face the nozzles **249a**, **249b** and **249c** (the gas supply holes **250a**, the gas supply holes **250b** and the gas supply hole **250c**) with the wafer **200** interposed therebetween when viewed from above. The exhaust port **231a** may be provided so as to extend upward from the lower portion toward the upper portion of the reaction tube **203** along a side wall of the reaction tube **203** (that is, along the wafer arrangement region). An exhaust pipe **231** is connected to the exhaust port **231a**. A vacuum pump **246** serving as a vacuum exhaust apparatus is connected to the exhaust pipe **231** through a pressure sensor **245** and an APC (Automatic Pressure Controller) valve **244**. The pressure sensor **245** serves as a pressure detector (pressure detection structure) to detect an inner pressure of the process chamber **201**, and the APC valve **244** serves as a pressure regulator (pressure adjusting structure). With the vacuum pump **246** in operation, the APC valve **244** may be opened or closed to vacuum-exhaust the process chamber **201** or stop the vacuum exhaust. With the vacuum pump **246** in operation, the inner pressure of the process chamber **201** may be adjusted by adjusting an opening degree of the APC valve **244** based on pressure information detected by the pressure sensor **245**. An exhauster (which is an exhaust structure or an exhaust system) is constituted mainly by the exhaust pipe **231**, the APC valve **244** and the pressure sensor **245**. The exhauster may further include the vacuum pump **246**.

(25) A seal cap **219** serving as a furnace opening lid capable of airtightly sealing (or closing) a lower end opening of the manifold **209** is provided under the manifold **209**. The seal cap **219** is capable of sealing the lower end opening of the manifold **209** when a boat **217** is rotatably supported in the process vessel. The seal cap **219** is made of a metal material such as SUS, and is of a disk shape. An O-ring **220b** serving as a seal is provided on an upper surface of the seal cap **219** so as to be in contact with the lower end of the manifold **209**. A rotator **267** configured to rotate the boat **217** described later is provided under the seal cap **219**. A rotating shaft **255** of the rotator **267** is connected to the boat **217** through the seal cap **219**. As the rotator **267** rotates the boat **217**, the wafers **200** accommodated in the boat **217** are rotated. The rotator **267** is provided with an annular groove (not shown), and is configured such that a purge gas is supplied through the annular groove to the vicinity of the rotating shaft **255**. As a result, it is possible to prevent the rotating shaft **255** from being corroded by the process gas in the process chamber **201**. The seal cap **219** is elevated or lowered in the vertical direction by a boat elevator **115** serving as an elevator provided outside the reaction tube **203**. The boat elevator **115** serves as a transfer device (which is a transfer structure or a transfer system) that loads the boat **217** and the wafers **200** accommodated in the boat **217** into the process chamber **201** or unloads the boat **217** and the wafers **200** accommodated in the boat **217** out of the process chamber **201** by elevating or lowering the seal cap **219**.

(26) A furnace opening shutter **219s** serving as a furnace opening lid capable of airtightly sealing (or closing) the lower end opening of the manifold **209** is provided under the manifold **209**. A furnace opening assembly is constituted mainly by the manifold **209** and the furnace opening shutter **219s**. The furnace opening shutter **219s** is configured to close the lower end opening of the manifold **209** when the seal cap **219** is lowered by the boat elevator **115** and the boat **217** is unloaded out of the process chamber **201**. For example, the furnace opening shutter **219s** is made of a metal material such as SUS, and is of a disk shape. An O-ring **220c** serving as a seal is provided on an upper surface of the furnace opening shutter **219s** so as to be in contact with the lower end of the manifold **209**. An opening and closing operation of the furnace opening shutter

219s such as an elevation operation and a rotation operation is controlled by a shutter opener/closer (which is a shutter opening/closing structure) **115s**. The furnace opening shutter **219s** is different from the seal cap **219** in that a film-forming quality is not considered for the furnace opening shutter **219s** and a mirror surface polishing for reducing particles is not performed on the furnace opening shutter **219s**. The furnace opening shutter **219s** will be described later in detail.

(27) The boat **217** serving as a substrate retainer is configured such that the wafers **200** (for example, 25 wafers to 200 wafers) are accommodated (or supported) in the vertical direction in the boat **217** while the wafers **200** are horizontally oriented with their centers aligned with one another with a predetermined interval therebetween in a multistage manner. For example, the boat **217** is made of a heat resistant material such as quartz and SiC. For example, a plurality of heat insulation plates **218** made of a heat resistant material such as quartz and SiC are supported at a lower portion of the boat **217** in a multistage manner.

(28) A temperature sensor **263** serving as a temperature detector is installed in the reaction tube **203**. A state of electric conduction to the heater **207** is adjusted based on temperature information detected by the temperature sensor **263** such that a desired temperature distribution of an inner temperature of the process chamber **201** can be obtained. The temperature sensor **263** is provided along the inner wall of the reaction tube **203**.

(29) As shown in FIG. 3, the controller **121** serving as a control device (control structure) is constituted by a computer including a CPU (Central Processing Unit) **121a**, a RAM (Random Access Memory) **121b**, a memory **121c** and an I/O port **121d**. The RAM **121b**, the memory **121c** and the I/O port **121d** may exchange data with the CPU **121a** through an internal bus **121e**. For example, an input/output device **122** constituted by a component such as a touch panel is connected to the controller **121**.

(30) The memory **121c** is configured by a component such as a flash memory, a hard disk drive (HDD) and a solid state drive (SSD). For example, a control program configured to control the operation of the substrate processing apparatus or a process recipe containing information on sequences and conditions of a substrate processing described later is readably stored in the memory **121c**. The process recipe is obtained by combining steps (sequences or processes) of the substrate processing described later such that the controller **121** can execute the steps to acquire a predetermined result, and functions as a program. Hereinafter, the process recipe and the control program may be collectively or individually referred to as a “program”. In addition, the process recipe may also be simply referred to as a “recipe”. Thus, in the present specification, the term “program” may refer to the recipe alone, may refer to the control program alone, or may refer to both of the recipe and the control program. The RAM **121b** functions as a memory area (work area) where a program or data read by the CPU **121a** is temporarily stored.

(31) The I/O port **121d** is connected to the above-described components such as the MFCs **241a** through **241h**, the valves **243a** through **243h**, the pressure sensor **245**, the APC valve **244**, the vacuum pump **246**, the temperature sensor **263**, the heater **207**, the rotator **267**, the boat elevator **115** and the shutter opener/closer **115s**.

(32) The CPU **121a** is configured to read the control program from the memory **121c** and execute the read control program. In addition, the CPU **121a** is configured to read the recipe from the memory **121c** in accordance with an operation command inputted from the input/output device **122**. According to the contents of the read recipe, the CPU **121a** may be configured to be capable of controlling various operations such as flow rate adjusting operations for various gases by the MFCs **241a** through **241h**, opening and closing operations of the valves **243a** through **243h**, an opening and closing operation of the APC valve **244**, a pressure adjusting operation by the APC valve **244** based on the pressure sensor **245**, a start and stop of the vacuum pump **246**, a temperature adjusting operation by the heater **207** based on the temperature sensor **263**, an operation of adjusting the rotation and the rotation speed of the boat **217** by the rotator **267**, an elevating and lowering operation of the boat **217** by the boat elevator **115** and an opening and

closing operation of the furnace opening shutter **219s** by the shutter opener/closer **115s**.

(33) The controller **121** may be embodied by installing the above-described program stored in an external memory **123** into a computer. For example, the external memory **123** may include a magnetic disk such as a hard disk drive (HDD), an optical disk such as a CD, a magneto-optical disk such as an MO and a semiconductor memory such as a USB memory and a solid state drive (SSD). The memory **121c** or the external memory **123** may be embodied by a non-transitory computer readable recording medium. Hereafter, the memory **121c** and the external memory **123** may be collectively or individually referred to as a recording medium. In the present specification, the term “recording medium” may refer to the memory **121c** alone, may refer to the external memory **123** alone, and may refer to both of the memory **121c** and the external memory **123**. Instead of the external memory **123**, a communication means such as the Internet and a dedicated line may be used for providing the program to the computer.

(34) Subsequently, the furnace opening shutter **219s** will be described with reference to FIG. **4**. The furnace opening shutter **219s** may include: a protective cover **219w** whose exposed surface is an inner surface of the furnace opening shutter **219s**, facing an inside of the manifold **209** and greater than the lower end opening of the manifold **209**; and a lid **219t** whose exposed surface is an outer surface of the furnace opening shutter **219s**, supporting the protective cover **219w**. In the above, the “outer surface of the furnace opening shutter **219s**” refers to a surface of the furnace opening shutter **219s** facing an outside of the process vessel when viewed from a furnace opening, and the “inner surface of the furnace opening shutter **219s**” refers to a surface of the furnace opening shutter **219s** facing an inside of the process vessel when viewed from the furnace opening. The protective cover **219w** is made of a material whose corrosion resistance is higher than the lid **219t**. For example, the protective cover **219w** is made of a material such as quartz glass, and may also be referred to as a “shutter glass” or a “quartz plate”. An outer peripheral surface (also referred to as a “second sealing surface”) of the furnace opening shutter **219s** faces a lower surface (also referred to as a “first sealing surface”) of the manifold **209**. The second sealing surface is located between an outer periphery “A” of the lid **219t** and an inner periphery “B” of the manifold **209**, as shown in FIGS. **4** and **5**, and is configured by an outer peripheral portion of the protective cover **219w** and an outer peripheral portion of the lid **219t**. A gap **219u** is provided between the lower surface of the manifold **209** and the second sealing surface. The O-ring **220c** is provided on the lid **219t** at an outermost portion of the second sealing surface, and a groove **219v** is formed on and along an entire circumference of the lid **219t** at a portion of the second sealing surface radially more inward than the outermost portion of the second sealing surface. The outer peripheral portion of the protective cover **219w** is provided on an upper surface of the lid **219t** and is located radially more inward than the groove **219v**.

(35) A lower flange of the manifold **209** is configured such that a hole **249i** serving as a gas introduction structure is provided at a position facing the groove **219v** so as to communicate with an inside of the process chamber **201**. A purge gas supply pipe **232i** through which the purge gas is supplied into the process chamber **201** is connected to the hole **249i**. A purge gas supply source (not shown) is connected to an upstream side of the purge gas supply pipe **232i** via a valve **243i** and an MFC **241i** serving as a gas flow rate controller at a location opposite to a connection location at which the purge gas supply pipe **232i** is connected to the hole **249i**. The purge gas is supplied through the hole **249i** to the groove **219v** located in the gap **219u** between the lower surface of the manifold **209** and the second sealing surface, passes through the gap **219u** between the lower surface of the manifold **209** and the outer peripheral portion of the protective cover **219w**, and is supplied to an inner peripheral portion of the protective cover **219w**. Preferably, the purge gas is supplied through the groove **219v** while the cleaning gas is supplied into the process vessel. A flow rate of the purge gas supplied through the groove **219v** is reduced when the cleaning gas is supplied into the process vessel as compared with a case where the cleaning gas is not supplied into the process vessel. An inner temperature of the process vessel when a cleaning process is performed

may be set to be higher than that of the process vessel when the wafer **200** is processed. The groove **219v** may be provided so as to face a lower surface of the lower flange of the manifold **209**. Further, the gap **219u** may be provided on and along the entire circumference of the lid **219t** or may be provided intermittently.

(36) The protective cover **219w** is large enough to extend under the manifold **209**, and the protective cover **219w** is purged along an entire circumference thereof via the groove **219v** provided on an outer peripheral portion of the lid **219t**. By purging the protective cover **219w** along the entire circumference thereof, it is possible to prevent (or suppress) the cleaning gas from entering a location between the lid **219t** and the protective cover **219w** (that is, a back surface of the protective cover **219w**). As a result, it is possible to prevent a corrosion of the lid **219t**.

(37) Further, the lid **219t** is provided with a flow path **219x** of cooling water serving as a cooling medium. The flow path **219x** is arranged at a central portion of the lid **219t**. For example, as shown in FIGS. **4** and **5**, the flow path **219x** includes: a first portion located about halfway between an outer periphery and a center of the lid **219t** and extending along a circumferential direction of the lid **219t**; and a second portion located radially more inward than the first portion and extending along the opposite circumferential direction of the lid **219t**. By supplying the cooling water through the flow path **219x**, it is possible to maintain a temperature of the lid **219t** within a predetermined temperature range (for example, within a range from 160° C. to 200° C.), inclusive.

(38) Subsequently, a cooling water system (which is a cooling water supply structure) configured to maintain the temperature of the lid **219t** within the predetermined temperature range will be described with reference to FIG. **6**. When a surface temperature of the furnace opening shutter **219s** is equal to or less than a predetermined temperature (for example, 150° C.), a cleaning reaction rate may decrease, the cleaning gas may be condensed, or a detachment of by-products may decrease. In order to maintain the surface temperature of the furnace opening shutter **219s** at a target temperature at which the cleaning process is effectively performed, a temperature control is performed by turning a supply of the cooling water on or off. In addition, the cooling water already used for other components (for example, the heater **207** and its peripheries) may be reused for the lid **219t**. Thereby, the surface temperature of the furnace opening shutter **219s** is maintained by using such warmed cooling water.

(39) Therefore, in order to detect a temperature of the cooling water in the flow path **219x**, a temperature sensor **301** is provided in the vicinity of the flow path **219x**. Although one temperature sensor **301** is illustrated in FIG. **6**, a plurality of temperature sensors including the temperature sensor **301** may be provided. In addition, a cooling medium supply pipe **311** through which the cooling water is supplied is connected to the heater **207**, and a discharge pipe **312** through which the cooling water warmed by cooling the heater **207** is discharged is connected to the heater **207** and a flow path switch **302**. A cooling medium supply pipe **313** through which the cooling water discharged from the flow path switch **302** is supplied is connected to an inlet of the flow path **219x** of the lid **219t**, and a discharge pipe **315** through which the cooling water warmed by cooling the lid **219t** is discharged is connected to an outlet of the flow path **219x** and a flow meter **303**. The flow path switch **302** switches a flow path of the cooling water based on the temperature sensor **301**. For example, when the temperature of the cooling water measured by the temperature sensor **301** is equal to or higher than a predetermined temperature, the cooling water is supplied to the furnace opening shutter **219s** via the cooling medium supply pipe **313**. When the temperature of the cooling water is lower than the predetermined temperature, the cooling water is discharged through a discharge pipe **314** such that a flow rate of the cooling water of the lid **219t** is reduced or the supply of the cooling water is stopped. For example, the flow meter **303** is configured as a Karman flow meter. The predetermined temperature is set to be equal to or less than a heat resistant temperature of the Karman flow meter, preferably 85° C.

(40) (2) Substrate Processing

(41) Hereinafter, an example of the substrate processing such as a film-forming process of forming

a predetermined film on the wafer **200** will be described with reference to FIG. 7. In the following descriptions, the operations of the components constituting the substrate processing apparatus are controlled by the controller **121**.

(42) According to the film-forming process of the present embodiments, a metal oxide film is formed on the wafer **200** by performing a cycle a predetermined number of times (at least once). For example, the cycle may include: a step **S941** of supplying the source gas to the wafer **200** in the process chamber **201**; a step **S942** of removing the source gas (residual gas) from the process chamber **201**; a step **S943** of supplying the reactive gas to the wafer **200** in the process chamber **201**; and a step **S944** of removing the reactive gas (residual gas) from the process chamber **201**. The steps **S941**, **S942**, **S943** and **S944** of the cycle are non-simultaneously performed.

(43) In the present specification, the term “wafer” may refer to “a wafer itself (a bare wafer)” or may refer to “a wafer and a stacked structure (aggregated structure) of a predetermined layer (or layers) or a film (or films) formed on a surface of the wafer”. Similarly, the term “a surface of a wafer” may refer to “a surface of a wafer itself” or may refer to “a surface of a predetermined layer or film formed on the wafer, that is, a top surface (uppermost surface) of the wafer as a stacked structure”. In the present specification, the term “substrate” and “wafer” may be used as substantially the same meaning. That is, the term “substrate” may be substituted by “wafer” and vice versa.

(44) <**S901**: Wafer Charging Step and Boat Loading Step>

(45) First, the wafers **200** are charged (transferred) into the boat **217** (wafer charging step). Then, the furnace opening shutter **219s** is moved by the shutter opener/closer **115s** to open the lower end opening of the manifold **209** (shutter opening step). Thereafter, as shown in FIG. 1, the boat **217** charged with the wafers **200** is elevated by the boat elevator **115** and loaded (transferred) into the process chamber **201** (boat loading step). With the boat **217** loaded, the seal cap **219** airtightly seals the lower end of the manifold **209** via the O-ring **220b**.

(46) <**S902**: Pressure Adjusting Step>

(47) Thereafter, the vacuum pump **246** vacuum-exhausts (decompresses and exhausts) the inner atmosphere of the process chamber **201** such that the inner pressure of the process chamber **201** in which the wafers **200** are accommodated reaches and is maintained at a desired pressure (vacuum degree) (pressure adjusting step). When the vacuum pump **246** vacuum-exhausts the inner atmosphere of the process chamber **201**, the inner pressure of the process chamber **201** is measured by the pressure sensor **245**, and the APC valve **244** is feedback-controlled based on the pressure information detected by the pressure sensor **245**. The vacuum pump **246** continuously vacuum-exhausts the inner atmosphere of the process chamber **201** until at least the processing of the wafer **200** (that is, from a temperature elevating step **S903** to a temperature lowering step **S905**) is completed. When the vacuum pump **246** vacuum-exhausts the inner atmosphere of the process chamber **201**, the valves **243e** through **243g** may be opened to continuously supply the inert gas whose flow rate is adjusted by the MFCs **241e** through **241g** to a small flow rate to a predetermined location in the process chamber **201**. The seal cap **219** distributes the purge gas supplied through the purge gas supply pipe **232i** to the rotator **267** such that the rotator **267** is purged, and a supply amount of the purge gas to purge the rotator **267** is about 100 sccm to 1,000 sccm.

(48) <**S903**: Temperature Elevating Step>

(49) The heater **207** heats the process chamber **201** such that the temperature of the wafer **200** in the process chamber **201** reaches and is maintained at a desired process temperature (temperature elevating step). When the heater **207** heats the process chamber **201**, the state of the electric conduction to the heater **207** is feedback-controlled based on the temperature information detected by the temperature sensor **263** such that the inner temperature of the process chamber **201** is stabilized at a desired temperature distribution. For example, the process temperature is set to be within a range from 100° C. to 650° C. When it is preferable to set the process temperature so as not to affect a parameter such as an impurity concentration in a channel region and a metal wiring,

for example, the process temperature is set to be within a range from 100° C. to 350° C. In addition, the rotation of the wafer **200** is started by the rotator **267**. The heater **207** continuously heats the wafer **200** in the process chamber **201** and the rotator **267** continuously rotates the wafer **200** until at least the processing of the wafer **200** is completed.

(50) <**S904: Film-Forming Step**>

(51) When the inner temperature of the process chamber **201** is stabilized at a predetermined process temperature, the film-forming step **S904** is performed by sequentially performing the following four sub-steps, that is, the steps **S941**, **S942**, **S943** and **S944**. During the film-forming step **S904**, the rotator **267** continuously rotates the boat **217** and the wafers **200** via the rotating shaft **255**.

(52) <**S941: Source Gas Supply Step**>

(53) In the source gas supply step **S941**, by supplying the source gas to the wafer **200** in the process chamber **201**, a first layer is formed on an outermost surface of the wafer **200**. Specifically, the valve **243b** is opened to supply the source gas into the gas supply pipe **232b**. A flow rate of the source gas supplied into the gas supply pipe **232b** is adjusted by the MFC **241b**. Then, the source gas whose flow rate is adjusted is supplied into a process region of the process chamber **201** through the gas supply holes **250b** of the nozzle **249b**, and is exhausted through the exhaust pipe **231** via the exhaust port **231a**. In the source gas supply step **S941**, simultaneously, the valve **243f** is opened to supply the inert gas into the gas supply pipe **232f**. A flow rate of the inert gas supplied into the gas supply pipe **232f** is adjusted by the MFC **241f**. Then, the inert gas whose flow rate is adjusted is supplied into the process region of the process chamber **201** together with the source gas through the gas supply holes **250b** of the nozzle **249b**, and is exhausted through the exhaust pipe **231** via the exhaust port **231a**. In addition, simultaneously, the inert gas is supplied into the process region of the process chamber **201** through the gas supply holes **250a** of the nozzle **249a** and the gas supply hole **250c** of the nozzle **249c**, and is exhausted through the exhaust pipe **231** via the exhaust port **231a**. In the source gas supply step **S941**, the controller **121** performs a constant pressure control by setting a first pressure as a target pressure. As the source gas, a gas containing a metal element such as aluminum (Al), zirconium (Zr), hafnium (Hf) and titanium (Ti) may be used.

(54) <**S942: Source Gas Exhaust Step**>

(55) After the first layer is formed, the valve **243b** is closed to stop the supply of the source gas into the process chamber **201**, and a pressure control is performed with the APC valve **244** fully opened. As a result, the inner atmosphere of the process chamber **201** is vacuum-exhausted to remove a residual gas such as the source gas in the process chamber **201** which did not react or which contributed to the formation of the first layer from the process chamber **201**. In the source gas exhaust step **S942**, with the valve **243f** open, the inert gas may be supplied into the process chamber **201** to further purge the residual gas. A flow rate of the purge gas (that is, the inert gas) through the nozzle **249b** is set such that a partial pressure of a low vapor pressure gas is lower than a saturated vapor pressure in an exhaust path, or such that a flow velocity of the gas is greater than a diffusion rate of the gas in the reaction tube **203**.

(56) <**S943: Reactive Gas Supply Step**>

(57) After the source gas exhaust step **S942** is completed, the reactive gas is supplied to the wafer **200** in the process chamber **201** (that is, to the first layer formed on the wafer **200**). In the reactive gas supply step **S943**, the reactive gas is thermally activated and then supplied to the wafer **200**. The thermally activated reactive gas reacts with at least a portion of the first layer formed on the wafer **200** in the source gas supply step **S941**. As a result, the first layer is modified (changed) into a second layer. In the reactive gas supply step **S943**, the valves **243a** and **243e** are controlled in the same manners as the valves **243b** and **243f** in the source gas supply step **S941**. Specifically, a flow rate of the reactive gas is adjusted by the MFC **241a**. The reactive gas whose flow rate is adjusted is then supplied into the process region of the process chamber **201** through the gas supply holes **250a** of the nozzle **249a**, and is exhausted through the exhaust pipe **231** via the exhaust port **231a**.

In addition, simultaneously, the inert gas is supplied into the process region of the process chamber **201** through the gas supply holes **250a** of the nozzle **249a**, the gas supply holes **250b** of the nozzle **249b** and the gas supply hole **250c** of the nozzle **249c**, and is exhausted through the exhaust pipe **231** via the exhaust port **231a**. In the reactive gas supply step **S943**, the controller **121** performs the constant pressure control by setting a second pressure as the target pressure. For example, the first pressure and the second pressure may be set to a pressure within a range of 100 Pa to 5,000 Pa, preferably within a range of 100 Pa to 500 Pa. As the reactive gas, for example, an oxygen-containing gas such as oxygen (O.sub.2) gas, ozone (O.sub.3) gas, plasma-excited O.sub.2 gas (O.sub.2*), a mixed gas of the O.sub.2 gas and hydrogen (H.sub.2) gas, water vapor (H.sub.2O gas), hydrogen peroxide (H.sub.2O.sub.2) gas, nitrous oxide (N.sub.2O) gas, nitrogen monoxide (NO) gas, nitrogen dioxide (NO.sub.2) gas, carbon monoxide (CO) gas and carbon dioxide (CO.sub.2) gas may be used.

(58) <**S944: Reactive Gas Exhaust Step**>

(59) After the second layer is formed, the valve **243a** is closed to stop the supply of the reactive gas into the process chamber **201**, and the constant pressure control (that is, a fully open control) by setting a zero (0) pressure as the target pressure is performed. As a result, the inner atmosphere of the process chamber **201** is vacuum-exhausted to remove a residual gas such as the reactive gas in the process chamber **201** which did not react or which contributed to the formation of the second layer from the process chamber **201**. In the reactive gas exhaust step **S944**, similar to the source gas exhaust step **S942**, a small amount of the inert gas may be supplied into the process chamber **201** as the purge gas. The ultimate pressure in the source gas exhaust step **S942** or the reactive gas exhaust step **S944** may be 100 Pa or less, preferably may be set to a pressure within a range of 10 Pa to 50 Pa. The inner pressure of the process chamber **201** in the source gas supply step **S941** or the reactive gas supply step **S943** may be different from that of the process chamber **201** in the source gas exhaust step **S942** or the reactive gas exhaust step **S944** by 10 times or more.

(60) <**S945: Performing Predetermined Number of Times**>

(61) By performing the cycle a predetermined number of times (n times) wherein the steps **S941** through **S944** described above are performed sequentially and non-simultaneously in this order, the film with a predetermined composition and a predetermined thickness is formed on the wafer **200**. Thicknesses of the first layer and the second layer formed in the steps **S941** and **S943**, respectively, may not be self-limiting. Therefore, in order to obtain a stable film quality, it is preferable that a concentration of the gas exposed to the wafer **200** and a supply time (time duration) of the gas exposed to the wafer **200** are precisely controlled with a high reproducibility. In addition, the steps **S941** and **S942** or the steps **S943** and **S944** may be performed (repeated) a plurality of times within the cycle.

(62) <**S905: Temperature Lowering Step**>

(63) In the temperature lowering step **S905**, the inner temperature of the process chamber **201** is gradually lowered when necessary, for example, by stopping the temperature elevating step **S903** which is continuously performed during the film-forming step **S904** or by re-setting the predetermined process temperature of the temperature elevating step **S903** to a lower temperature.

(64) <**S906: Purge Step and Returning to Atmospheric Pressure Step**>

(65) After the film-forming step **S904** is completed, the inert gas is supplied into the process chamber **201** through each of the nozzles **249a**, **249b** and **249c**, and then is exhausted through the exhaust port **231a**. The inert gas supplied through the nozzles **249a**, **249b** and **249c** serves as the purge gas. Thereby, the process chamber **201** is purged with the inert gas such that the residual gas or reaction by-products remaining in the process chamber **201** are removed from the process chamber **201** (purge step). Thereafter, the inner atmosphere of the process chamber **201** is replaced with the inert gas (substitution by inert gas), and the inner pressure of the process chamber **201** is returned to the atmospheric pressure (returning to atmospheric pressure step).

(66) <**S907: Boat Unloading Step and Wafer Discharging Step**>

(67) Thereafter, the seal cap **219** is lowered by the boat elevator **115** and the lower end of the manifold **209** is opened. Then, the boat **217** with the processed wafers **200** charged therein is unloaded (transferred) out of the reaction tube **203** through the lower end of the manifold **209** (boat unloading step). After the boat **217** is unloaded, the furnace opening shutter **219s** is moved. Thereby, the lower end opening of the manifold **209** is sealed by the furnace opening shutter **219s** through the O-ring **220c** (shutter closing step). The processed wafers **200** are taken out of the reaction tube **203**, and then discharged from the boat **217** (wafer discharging step).

(68) <**S908**: Pressure Adjusting Step>

(69) Thereafter, the vacuum pump **246** vacuum-exhausts (decompresses and exhausts) the inner atmosphere of the process chamber **201** such that the inner pressure of the process chamber **201** reaches and is maintained at a desired pressure (vacuum degree) (pressure adjusting step). When the vacuum pump **246** vacuum-exhausts the inner atmosphere of the process chamber **201**, the inner pressure of the process chamber **201** is measured by the pressure sensor **245**, and the APC valve **244** is feedback-controlled based on the pressure information detected by the pressure sensor **245**. The vacuum pump **246** continuously vacuum-exhausts the inner atmosphere of the process chamber **201** until at least a cleaning step **S910** is completed. While the vacuum pump **246** continuously vacuum-exhausts the inner atmosphere of the process chamber **201**, the valves **243e** through **243g** may be opened to supply a small flow rate of the inert gas.

(70) <**S909**: Temperature Elevating Step>

(71) The heater **207** heats the process chamber **201** such that the inner temperature of the process chamber **201** reaches and is maintained at a desired process temperature (temperature elevating step). When the heater **207** heats the process chamber **201**, the state of the electric conduction to the heater **207** is feedback-controlled based on the temperature information detected by the temperature sensor **263** such that a desired temperature distribution of the inner temperature of the process chamber **201** can be obtained. The heater **207** continuously heats the process chamber **201** until at least the cleaning step **S910** is completed.

(72) <**S910**: Cleaning Step>

(73) When the inner temperature of the process chamber **201** is stabilized at a predetermined process temperature, the cleaning gas is supplied into the process chamber **201**. Specifically, at least one among the valve **243c** and the valve **243d** is opened. A flow rate of the cleaning gas supplied into at least one among the gas supply pipe **232c** and the gas supply pipe **232d** is adjusted by at least one among the MFCs **241c** and **241d**. Then, the cleaning gas whose flow rate is adjusted is supplied into the process chamber **201** through at least one among the gas supply hole **250c** of the nozzle **249c** and the gas supply holes **250a** of the nozzle **249a**, and is exhausted through the exhaust pipe **231** via the exhaust port **231a**. In the cleaning step **S910**, the controller **121** performs the constant pressure control by setting a third pressure as the target pressure. The APC valve **244** may be nearly fully closed until the inner pressure of the process chamber **201** is elevated to the third pressure. Further, the inert gas supplied into the process chamber **201** may be reduced as much as possible.

(74) The cleaning gas is a gas containing a halogen atom. Various compounds described below may be used as the cleaning gas. As a compound containing a carbonyl group and a halogen atom in a molecule, a compound such as carbonyl fluoride (COF.sub.2), carbonyl chloride (COCl.sub.2), carbonyl bromide (COBr.sub.2), carbonyl chloride fluoride (COClF), carbonyl bromide fluoride (COBrF) and carbonyl iodide fluoride (COIF) may be used. Further, the number of carbonyl groups in the molecule is not limited to one (1), and a compound containing two carbonyl groups such as oxalyl fluoride (C.sub.2O.sub.2F.sub.2), oxalyl chloride (C.sub.2O.sub.2Cl.sub.2) and oxalyl bromide (C.sub.2O.sub.2Br.sub.2) may also be used. As a compound containing a thionyl group and the halogen atom in the molecule, a compound such as thionyl fluoride (SOF.sub.2), thionyl chloride (SOCl.sub.2), thionyl bromide (SOBr.sub.2), thionyl tetrafluoride (SOF.sub.4) and thionyl chloride bromide (SOClBr) may be used. As a compound containing a sulfonyl group and the

halogen atom in the molecule, a compound such as sulfuryl chloride (SO₂Cl₂), sulfuryl chloride fluoride (SO₂ClF) and sulfuryl bromide fluoride (SO₂BrF) may be used. As a compound containing a nitrosyl group and the halogen atom in the molecule, a compound such as nitrosyl fluoride (NOF), nitrosyl chloride (NOCl), nitrosyl dichloride (NOCl₂), nitrosyl trichloride (NOCl₃), nitrosyl bromide (NOBr), nitrosyl dibromide (NOBr₂) and nitrosyl tribromide (NOBr₃) may be used. As a compound containing a nitrile group and the halogen atom in the molecule, a compound such as nitrile fluoride (NO₂F), nitrile bromide (NO₂Br) and nitrile chloride (NO₂Cl) may be used.

(75) An oxide containing an element such as aluminum (Al), hafnium (Hf) and zirconium (Zr) formed in the film-forming step **S904** is known as a high-k material. However, the oxide containing an element such as aluminum (Al), hafnium (Hf) and zirconium (Zr) is difficult to etch. When the oxide described above is converted to a chloride or a bromide whose vapor pressure is high, it is possible to etch the converted oxide without using the plasma. A cleaning process temperature (that is, the process temperature in the cleaning step **S910**) is set to be higher than a film-forming process temperature (that is, the process temperature in the film-forming step **S904**). For example, the cleaning process temperature may be set to a temperature within a range of 400° C. to 750° C. For example, the third pressure may be set to a pressure within a range of about 10 kPa to 16,600 kPa. In the cleaning step **S910**, the controller **121** can maintain a temperature on the inner surface of the furnace opening shutter **219s** at about the target temperature by the following control. That is, when the temperature of the cooling water measured by the temperature sensor **301** exceeds a first predetermined temperature, the supply of the cooling water to the flow path **219x** is forcibly turned on, and after a predetermined time “t1” or after a predetermined amount “w” of the cooling water has flowed, the supply of the cooling water to the flow path **219x** is turned off. An amount of cooling water supplied through the flow path **219x** is obtained by integrating values of the flow meter **303**, and the predetermined amount w is determined corresponding to a volume of the flow path **219x**. After the supply of the cooling water to the flow path **219x** is turned off, the supply of the cooling water to the flow path **219x** is maintained turned off for a predetermined time “t2” regardless of a temperature of the temperature sensor **301**. Thereafter, it is waited until the temperature of the cooling water exceeds the first predetermined temperature again. In addition, a second temperature sensor (not shown) configured to measure a temperature on an inner surface of the lid **219t** may be added. In such a case, when a temperature measured by the second temperature sensor exceeds a second predetermined temperature, the supply of the cooling water to the flow path **219x** is forcibly turned on. The control using the temperature sensor **301** and the control using the second temperature sensor may be performed in parallel.

(76) <**S911**: Purge Step and Returning to Atmospheric Pressure Step>

(77) After the cleaning step **S910** is completed, the inert gas is supplied into the process chamber **201** through each of the nozzles **249a**, **249b** and **249c**, and then is exhausted through the exhaust port **231a**. The inert gas supplied through the nozzles **249a**, **249b** and **249c** serves as the purge gas. Thereby, the process chamber **201** is purged with the inert gas such that the residual gas or the reaction by-products remaining in the process chamber **201** are removed from the process chamber **201** (after-purge step). Thereafter, the inner atmosphere of the process chamber **201** is replaced with the inert gas (substitution by inert gas), and the inner pressure of the process chamber **201** is returned to the atmospheric pressure (returning to atmospheric pressure step).

(78) <Other Embodiments>

(79) While the technique of the present disclosure is described in detail by way of the embodiments described above, the technique of the present disclosure is not limited thereto. The technique of the present disclosure may be modified in various ways without departing from the scope thereof. Those skilled in the art may widely apply the embodiments described above to a heat treatment process of the substrate under a depressurized state. For example, the technique of the present disclosure is not limited to a hot wall type reaction tube, and may be applied to a cold wall type

reaction tube by using a lamp heating or induction heating. For example, the technique of the present disclosure may be applied to various types of reaction tubes such as a single tube type reaction tube shown in FIG. 1, a reaction tube with a buffer (duct) and a double tube type reaction tube.

(80) The film-forming process is not limited to that performed on the substrate such as the wafer. For example, the film-forming process may be performed on an inner wall of a chamber (process vessel) so as to protect the chamber and to prevent the particles from being generated. The furnace opening shutter according to the technique of the present disclosure may be used for coating (or re-coating) and removing an existing protective film before coating.

(81) According to some embodiments of the present disclosure, it is possible to reduce the corrosion of the furnace opening shutter.

Claims

1. A substrate processing apparatus comprising: a process vessel provided with an opening through which a substrate retainer accommodating a substrate is transferred and a first sealing surface around the opening; and a furnace opening shutter provided with a second sealing surface facing the first sealing surface, and capable of closing the opening in a state where the substrate retainer is not accommodated in the process vessel, wherein the furnace opening shutter comprises: a protective cover corresponding to an inner surface of the furnace opening shutter, facing the process vessel and greater than the opening; and a lid corresponding to an outer surface of the furnace opening shutter and supporting the protective cover, wherein the second sealing surface is constituted by an outer peripheral portion of the protective cover and the lid, and a gap is formed between the first sealing surface and the second sealing surface for a purge gas to be supplied thereto from a location radially more outward than the outer peripheral portion of the protective cover.
2. The substrate processing apparatus of claim 1, wherein the process vessel or the furnace opening shutter is provided with a groove through which the purge gas is supplied to the gap from the location radially more outward than the outer peripheral portion of the protective cover, and the protective cover is provided with an inner peripheral portion such that the purge gas after passing through the gap is capable of being supplied to the inner peripheral portion of the protective cover.
3. The substrate processing apparatus of claim 1, wherein the lid comprises a flow path of a cooling medium for a temperature of the lid to be maintained within a predetermined temperature range.
4. The substrate processing apparatus of claim 1, wherein the lid comprises: a flow path of a cooling medium capable of maintaining a temperature of the lid within a predetermined temperature range; and a temperature sensor configured to detect a temperature of the cooling medium in the flow path.
5. The substrate processing apparatus of claim 4, wherein the lid further comprises a flow path switch configured to switch the flow path of the cooling medium such that the cooling medium is supplied to the flow path when the temperature of the cooling medium is equal to or higher than a predetermined temperature, and a supply of the cooling medium is reduced or stopped when the temperature of the cooling medium is lower than the predetermined temperature.
6. The substrate processing apparatus of claim 4, further comprising a heater configured to heat the substrate, wherein the cooling medium after cooling the heater or its peripheries is supplied to the flow path.
7. The substrate processing apparatus of claim 4, wherein the flow path is provided at a central portion of the lid.
8. The substrate processing apparatus of claim 2, further comprising a purge gas supply pipe configured to supply the purge gas through the groove while a cleaning gas is being supplied into the process vessel.

9. The substrate processing apparatus of claim 8, further comprising a heater configured to heat the substrate such that a process temperature is higher when the cleaning gas is supplied than when the substrate is processed.
10. The substrate processing apparatus of claim 2, further comprising a purge gas supply pipe and a flow rate controller configured to supply the purge gas through the groove such that a flow rate of the purge gas supplied through the groove is smaller when a cleaning gas is supplied into the process vessel than when the cleaning gas is not supplied into the process vessel.
11. The substrate processing apparatus of claim 1, further comprising: a seal cap capable of closing the opening in a state where the substrate retainer is rotatable and supported in the process vessel; and a rotator provided on the seal cap to rotate the substrate retainer, wherein the rotator is configured such that the purge gas is supplied to a vicinity of a rotating shaft of the rotator through an annular groove.
12. The substrate processing apparatus of claim 4, wherein the predetermined temperature range is within a range from 160° C. to 200° C., inclusive, and the cooling medium comprises water.
13. A furnace opening assembly comprising: a manifold constituting a furnace opening of a process vessel, and provided with an opening through which a substrate retainer accommodating a substrate is transferred and a first sealing surface around the opening; and a furnace opening shutter provided with a second sealing surface facing the first sealing surface, and capable of closing the opening in a state where the substrate retainer is not accommodated in the process vessel, wherein the furnace opening shutter comprises: a protective cover provided corresponding to an inner surface of the furnace opening shutter facing the process vessel and greater than the opening; and a lid provided corresponding to an outer surface of the furnace opening shutter and supporting the protective cover, wherein the second sealing surface is constituted by an outer peripheral portion of the protective cover and the lid, and a gap is formed between the first sealing surface and the second sealing surface for a purge gas to be supplied thereto from a location radially more outward than the outer peripheral portion of the protective cover.
14. The substrate processing apparatus of claim 2, further comprising: a gas introduction hole penetrating a lower flange of a manifold and the opening on the first sealing surface at a position facing the groove that is provided at the lid.
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